



AI-Powered Drug Repurposing Discovery

Progress Report — AAI 595

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March 2026

What is BioLink?

A drug repurposing discovery tool that uses knowledge graph AI to find new therapeutic uses for existing drugs.

0.947
AUC-ROC

7,164
Drugs Scored



Patient types
a disease



AI maps to
medical entity



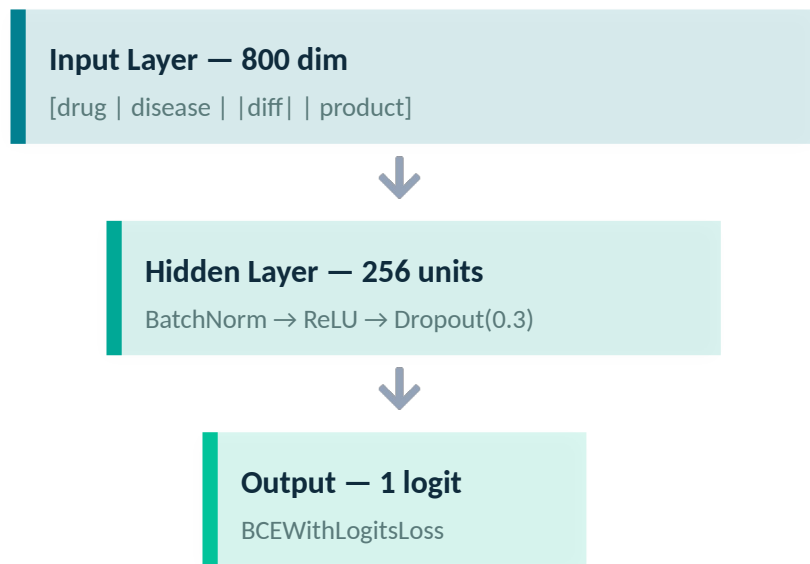
Model scores
7,164 drugs



Ranked results
with evidence

Model Architecture

MLP Network



Embeddings

BioWordVec

Pretrained on PubMed + MIMIC-III clinical notes
200-dimensional vectors
Tokenize → lookup → mean-pool per entity

Training Config

Data Source	CTD (Comparative Toxicogenomics DB)
Pairs	~78K (1:1 pos/neg ratio)
Split	70% train / 15% val / 15% test
Optimizer	Adam (lr = 1e-3)
Epochs	10
Batch Size	256
Seed	42

Performance & Calibration

Model Performance

0.947

AUC-ROC

0.949

Avg Precision

87.6%

Accuracy

Precision@K

@10

1.000

@50

1.000

@100

1.000

Top-ranked predictions are highly reliable

Temperature Calibration

Why calibrate?

Raw model outputs are overconfident.
A predicted 90% should mean the model
is right 90% of the time.

$T = 1.384$

`calibrated_prob = sigmoid(logit / T)`

Confidence Tiers

Strong

$\geq 80\%$

Moderate

50-80%

Speculative

$< 50\%$

System Architecture

Intent Mapping	Claude API resolves free-text to CTD disease entity
MLP Inference	BioWordVec embeddings + trained model scores all drugs
Calibration	Temperature scaling (T=1.38) for honest probs
Enrichment	PubMed evidence counts + FDA approval status
Explanation	Claude generates plain-English summaries

Tech Stack

- Python + PyTorch
- BioWordVec (200-dim)
- Claude API (Sonnet)
- Streamlit UI
- PubMed + OpenFDA APIs
- Hugging Face Spaces

Build Progress



Data assets & val set



Core model (MLP + BioWordVec)



Temperature calibration



Intent mapper (Claude API)



End-to-end inference pipeline



UI design mockups (Google Stitch)



Enrichment layer (PubMed + FDA)



Explanation layer (Claude)



Streamlit UI build



Tests + deployment

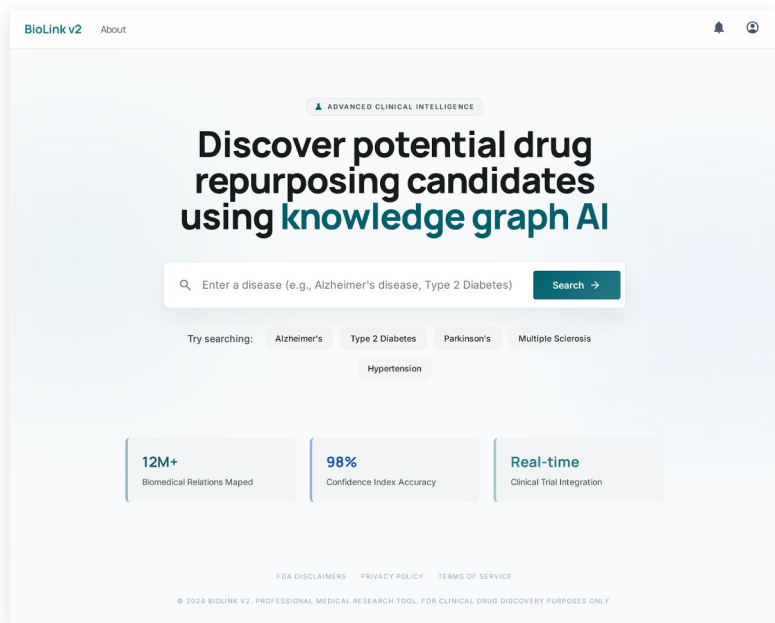
60% Complete



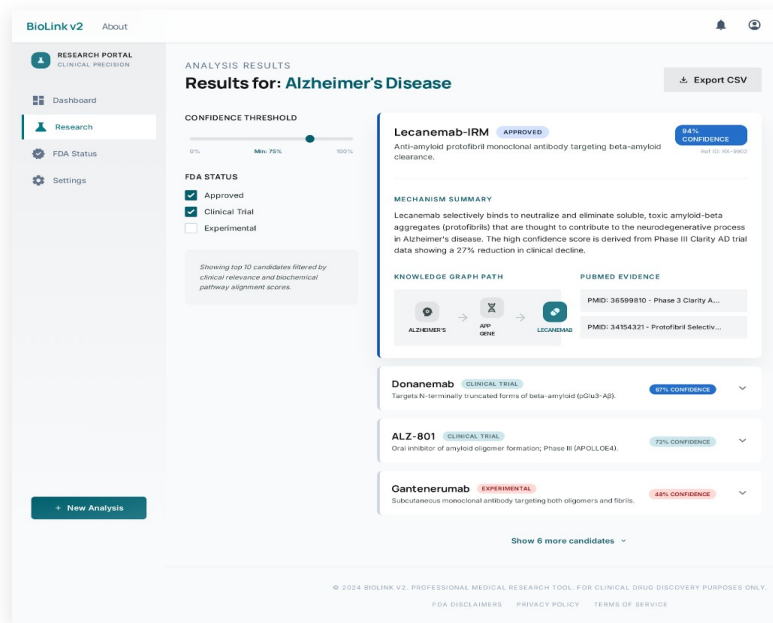
UI Design Preview

Designed in Google Stitch — mockups guide our Streamlit build

Search



Results



Team & Next Steps



Dustin Haggett

Lead — Core pipeline,
calibration, architecture



Kera Prosper

Explanation layer,
tests



Mia Braun

Enrichment layer
(PubMed + FDA)

Immediate Next Steps

- Build enrichment layer — async PubMed & OpenFDA lookups
- Build explanation layer — Claude-generated plain-English summaries
- Implement Streamlit UI from Google Stitch mockups
- Unit tests with mocked APIs, then deploy to Hugging Face Spaces



Repo is public with branch protection — all contributions via PRs with code review

Questions?

github.com/dustinhaggett/biolink-v2

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